Project Methodology Roadmap

Al Safa 2 Park — How Each Phase Produces Its Answer & Feeds the Next

This document shows the method — how the whole project was built, step by step. Each phase takes the previous phase's output as its input, so the design is traceable end-to-end: every decision leads back to real data. Read it to understand the logic of how we got from 'an empty site' to 'the final 12 submission files.'

The Chain — How Each Phase Feeds the Next

Phase	Input (from prev.)	What We Do (method)	Output (the answer)
1. Understand Site	Competition files (brief, DWG, map, manual)	Pull & compute REAL data: 8,760-hr solar (pvlib), sourced climate (Dubai Met Office), catchment (Dubai Stats Center), wind, transit. Python + pandas.	Existing Conditions Knowledge Base + SWOT
2. Define Problems	Phase 1 Knowledge Base (weaknesses, gaps)	Score every problem 1-5 on Evidence/Impact/Reach/Urgency with a weighted Python model; rank them.	Ranked problem list (P1 heat = 5.0/CRITICAL)
3. Set Objectives	Phase 2 ranked problems	Turn each top problem into a measurable objective + success metric.	Vision, mission, 5 objectives, 5 metrics
4. Choose Concept	Phase 3 objectives	Generate 3 concepts; score each on a weighted matrix (function, UX, sustainability, feasibility, innovation).	Winner: 'Shaded Spine' (7.85/10)
5. Master Plan	Phase 4 chosen concept	Draw the concept as REAL scaled geometry in Python — 13 zones summing to exactly 15,000 m2 + circulation.	Master plan + circulation diagram + area schedule
6. Detailed Design	Phase 5 zone geometry	Assign REAL native species per zone, place 131 trees, draw section + elevations to scale.	Planting plan, materials, section, elevations
7. Prove Performance	Phase 5 geometry + Phase 6 planting	COMPUTE everything: shade over 8,760 hrs, water budget (real Ghaf data), build cost (real Dubai rates), carbon, Heat-Index comfort, annual O&M.	99.2% shade, ~AED18.6M build, +3 comfort months, 2.1t CO2/yr, ~AED2M/yr O&M
8. User Experience	Phase 1 catchment + Phase 5 zones	Build personas from the real population; map their journeys through the actual zones.	5 personas, daily/seasonal use, journey maps
9. AI + Visualize	All prior phases	Document the AI method for every phase; generate renders (aerial day/night, eye-level) + 2 presentation boards.	AI methodology report + all visuals + boards
10. Package	Everything above	Auto-compile each output into its correct upload folder; QA against the brief; build the master report + checklist.	The final 12 submission files, verified

Worked Example — How Phase 1 Produces One Answer

Take the single most important finding, so you can see the method concretely:

- **Question:** "Does the existing park have usable shade in summer?"
- **Data pulled (real):** the site's exact latitude/longitude → fed to pvlib (NASA/NREL solar algorithm) to compute the sun's exact position for all 8,760 hours of the year.
- **Analysis (Python):** for the summer solstice at noon, the sun sits ~88° above the horizon → shadow length = object height ÷ tan(88°) ≈ almost zero.
- **Answer:** "No — a normal tree casts a <0.5m shadow at summer noon, so the site is unshaded when it's hottest." This becomes Problem P1 in Phase 2.

- **How it feeds forward:** P1 scored 5.0/CRITICAL (Phase 2) → became the #1 objective (Phase 3) → drove the 'Shaded Spine' concept (Phase 4) → which Phase 7 then PROVED delivers 99.2% shade. Full circle, all from one real data pull.

Worked Example — How Phase 2 Produces Its Answer

- **Question:** "Which problem should the design solve first?"
- **Data used:** the findings from all 13 Phase 1 sub-analyses.
- **Analysis (Python):** each candidate problem is scored 1-5 on four criteria — Evidence (25%), Impact (30%), Reach (25%), Urgency (20%) — and a weighted total is computed.
- **Answer:** a ranked list. P1 (summer heat) = 5.00 CRITICAL; P2 (accessibility) = 4.30 CRITICAL; then the rest. The design then addresses problems in THIS computed order.
- **Why it matters:** priorities are calculated from evidence, not guessed — which is exactly what a jury wants to see.

The Golden Rule Across All Phases

Every number is one of three things, always labelled: (a) computed from real astronomy/geometry, (b) pulled from a named real dataset with a date, or (c) a clearly-flagged assumption. Real data gaps (the exact DWG boundary, on-site soil/noise) are stated honestly, never invented. That integrity is what makes the whole chain credible.

Where to Find Each Phase's Work

Each phase lives in its numbered folder (01–10). Inside each: the analysis outputs, the PDF report, and a `_scripts/` folder with the exact Python code that generated it. The `_FINAL_DELIVERABLES/` folder gathers all the finished reports and visuals in one place. Start from `START_HERE.md` at the top level.